Prime-field word coefficient update 2026-10-31

T-1149. Producer proof packet; independent source-fidelity review pending.

Checked update and hypotheses

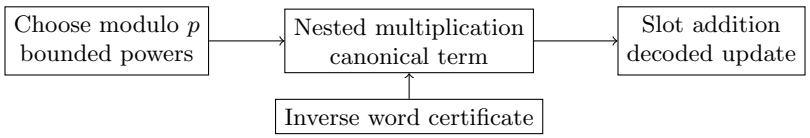
The word output remains canonical and decodes to

$$o + c\binom{k}{j} a^j b^{k-j} s^{-1}.$$

Hypotheses: prime $1 < p < 2^{31}$; canonical input words o, c, a, b, s, i_w; $0 \le j \le k \le 5$; $s \neq 0$; and the checkable word certificate

$$(s \cdot i_w) \bmod p = 1.$$

Bounded multiplication decoding proves that the certificate supplies the field inverse. This verifies the supplied word, not the inversion generator.



Supporting proofs and controls

The source-style binary power loop unrolls symbolically for exponents zero through five; three shifts drain the exponent. Bounded choose-word reduction preserves its residue-class value. Multiplication follows the source parenthesization; addition reuses the bounded word theorem.

Nine main declarations and seventeen controls receive separate named axiom readback. With $p = 17, a = 2, b = 3, c = 4, s = 2, i_w = 9, k = 2, j = 1$, the term is 7, agreeing with $24 \bmod 17$; prior slot 10 updates to 0. Controls cover zero exponent/base, $(-1)^5$, reduced choose, invalid indices and wrong/zero/noncanonical inverse certificates.

Inverse word 26 passes the product test for scale 2 modulo 17 but fails canonicality. For the excluded input 2^{63} times 2 modulo 5, the natural result is 1 and the UInt64 result is 0. These controls retain the input-bound distinction.

Source and evidence boundary

The source mapping is odd-characteristic $n = 1$ arithmetic without diagnostic tables. `Field::inv` rejects zero and calls `bi`; its general `modpow` uses u128 products. The arbitrary-exponent inversion generator, literal vector/compiler behavior and geometric class transport remain separate bindings. Exact excerpts and source hashes accompany this packet.

One main attempt needed an explicit import of the existing nonzero-decoding lemma. The first controls needed predicate unfolding for decidability; corrected checks passed. Native exits, unchanged bindings and axiom readback are retained with all attempts. Existing compiled dependencies remain in the trust boundary.

References: existing UInt64Residue/CrossScaling/WordAdd11 proofs and Mathlib ZMod.Basic. No scientific campaign, curve enumeration, timing or cost ratio. Cohort and campaign limits unchanged. Diagram shows proof dependencies.